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AIR HORNS AND METHOD OF USE

Abstract

Air horns and methods for their operation. Such an air horn includes a main body having a cavity, a base, an actuator housing, and a diaphragm housing and bell fluidically connected to the cavity and to a channel within the main body. The actuator housing has a wall, a cavity surrounded by the wall, and standoffs within the cavity that define first and second sets of gaps between pairs of the standoffs. An actuator is coupled to the actuator housing so as to have first and second rotational positions. The actuator includes a shaft received in the cavity and standoff guides that individually receive pairs of the standoffs. The standoffs and the standoff guides interact to only permit axial movement of the actuator relative to the actuator housing when the actuator is in the first and second rotational positions and not at any rotational position therebetween.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/559,016 filed Feb. 28, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention generally relates to air horns for use in combination with a container of a compressed gas. More particularly, the invention relates to air horns that include a lockable push-button actuator that is configured to prevent accidental discharge of compressed gas from a container.

[0003] Air horns are generally types of devices that use compressed air or other gas to produce an audible sound, as a nonlimiting example, a sound sufficiently loud to discourage, startle, or frighten an animal. Existing air horns are typically equipped with an actuating mechanism through which air is discharged to produce the sound. However, such actuating mechanisms are often not particularly well configured to prevent being unintentionally actuated, even if the actuating mechanism is in an “off” position.

[0004] It would be desirable if air horns were available that provide a more secure means for preventing an accidental discharge and yet easily operated to produce an audible sound when needed.

BRIEF SUMMARY OF THE INVENTION

[0005] The intent of this section of the specification is to briefly indicate the nature and substance of the invention, as opposed to an exhaustive statement of all subject matter and aspects of the invention. Therefore, while this section identifies subject matter recited in the claims, additional subject matter and aspects relating to the invention are set forth in other sections of the specification, particularly the detailed description, as well as any drawings.

[0006] The present invention provides, but is not limited to, to air horns for use with containers containing a compressed gas, and methods for their operation.

[0007] According to a nonlimiting aspect, an air horn is provided that is configured to be releasably connected to a container of compressed gas. The air horn includes a main body having a cavity, a channel wall within the cavity, and a channel defined and surrounded by the channel wall. The air horn further has a base having a wall extending into the main body, a diaphragm housing connected a first end of the main body and fluidically connected to the cavity and the channel of the main body, and a bell fluidly connected to the channel of the main body. The base has an inlet port of the air horn defined by the wall, an orifice disposed in the wall to fluidically connect the inlet port with the cavity of the main body, and connecting means for releasably connecting the air horn to a container of compressed gas. An actuator housing has a wall connected to the main body, a cavity surrounded by the wall of the actuator housing, and standoffs within the cavity of the actuator housing that define a first set of gaps between a first pair of the standoffs and a second set of gaps between a second pair of the standoffs. A pin is surrounded by the standoffs and adapted to axially reciprocate within a bore within the main body. An actuator is coupled to the actuator housing so as to have rotational orientations that include a first rotational position and in a second rotational position. The actuator includes a shaft received in the cavity of the actuator housing and standoff guides defined within the shaft that individually receive pairs of the standoffs. The standoffs and the standoff guides interact to only permit axial movement of the actuator relative to the actuator housing when the actuator is in the first rotational position and in the second rotational position and not at any rotational position therebetween. The first rotational position corresponds to an OFF position of the actuator and the second rotational position corresponds to an ON position of the actuator.

[0008] Another aspect of the invention is a method of operating an air horn as described above. The method includes releasably connecting the air horn with the connecting means to a container of compressed gas, positioning the actuator in the first rotational position to prevent actuation of the actuator and prevent release of the compressed gas from the container, and rotating the actuator to the second rotational position to permit actuation of the actuator and permit release of the compressed gas from the container.

[0009] Technical aspects of air horns as described above preferably include being equipped with an actuator configured to prevent or at least reduce the risk of intentionally activating the air horn when the actuator is in an off position or when a non-axial force is applied to the actuator.

[0010] These and other aspects, arrangements, features, and/or technical effects will become apparent upon detailed inspection of the figures and the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIGS. 1 and 2 are perspective side views of an air horn of the present invention mounted to a container of compressed gas.

[0012] FIGS. 3 and 4 schematically represent bottom and top perspective views, respectively, of the air horn of FIGS. 1 and 2.

[0013] FIG. 5 schematically represents a partially exploded top perspective view of the air horn of FIGS. 1 and 2.

[0014] FIG. 6 schematically represents a perspective end view of the air horn of FIGS. 1 and 2 with a diaphragm housing thereof removed.

[0015] FIG. 7 schematically represents a partial top perspective view of the air horn of FIGS. 1 and 2 showing an actuator thereof in an OFF position.

[0016] FIG. 8 schematically represents a detail of FIG. 7 with the actuator depicted as translucent to reveal internal structures of the actuator and an actuator housing of the air horn.

[0017] FIG. 9 schematically represents a partial top perspective view of the air horn of FIGS. 1 and 2 showing the actuator thereof in an ON position.

[0018] FIG. 10 schematically represents a detail of FIG. 9 with the actuator depicted as translucent to reveal internal structures of the actuator and the actuator housing.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The intended purpose of the following detailed description of the invention and the phraseology and terminology employed therein is to describe what is shown in the drawings, which include the depiction of and/or relate to one or more nonlimiting embodiments of the invention, and to describe certain but not all aspects of what is depicted in the drawings, including the embodiment(s) depicted in the drawings. The following detailed description also describes certain investigations relating to the embodiment(s) depicted in the drawings, and identifies certain but not all alternatives of the embodiment(s) depicted in the drawings. As nonlimiting examples, the invention encompasses additional or alternative embodiments in which one or more features or aspects shown and/or described as part of a particular embodiment could be eliminated. Therefore, the appended claims, and not the detailed description, are intended to recite particularly point out subject matter regarded to be aspects of the invention, including certain but not necessarily all of the aspects and alternatives described in the detailed description.

[0020] FIGS. 1 and 2 represent an air horn 20 of the present invention connected to a canister 10 that contains a compressed gas, as a nonlimiting example, air. The air horn 20 is operable to cause compressed gas from the canister 10 to pass through the air horn 20 to generate an audible sound. In some cases the audible sound may be at a sufficiently high decibel to be effect to discourage, startle, or frighten an animal or individual that may pose a threat to the user of the air horn 20,

though it is also within the scope of the invention that the audible sound could be at lower decibels and/or used for other purposes, as a nonlimiting example, as a signaling device on a boat or outdoors, which can be useful in the event of an emergency. Though the term “air” will be used herein to refer to the device represented in the drawings, use of this term is a matter of convenience and it should be understood that the air horn **20** is not limited to using air to produce an audible sound.

[0021] FIGS. **1** and **2** represent the air horn **20** as generally including a main body **22**, a base **24** extending downward from the main body **22**, a diaphragm housing **26** secured to a first side of the main body **22**, a bell **28** disposed at a second side of the main body **22** opposite the diaphragm housing **26**, an actuator housing **30** extending upward from the main body **22** opposite the base **24**, and an actuator **32** mounted to the actuator housing **30**. The base **24**, bell **28**, and actuator housing **30** may be integrally formed with the main body **22**, such as by injection molding, or formed separately and assembled together. As known in the art, the diaphragm housing **26** contains a diaphragm (not shown) that converts the energy of the compressed gas flowing from the canister **10** into sound waves that are emitted through the bell **28**. As shown in FIGS. **1** and **2**, the bell **28** is in the shape of a catenoid bell.

[0022] FIG. **3** schematically represents a lower side **40** of the air horn **20** and the base **24** protruding from the main body **22**. The canister **10** is removed to expose an interior cavity **62** of the base **24**. As shown, the base **24** has an annular or tubular-shaped wall **48** that defines an outer sidewall **52** at the exterior of the base **24**, an inner sidewall **54** within the cavity **62** of the base **24**, and a lip or rim **60** therebetween at a distal end of the base **24**. Connection means (not shown) such as threads may be provided within the inner sidewall **54** for connecting the air horn **20** to the canister **10**. A boss **56** within the cavity **62** defines a surface that is substantially planar and parallel to the rim **60** of the base **24**. The boss **56** is located within the cavity **62** to provide a bearing surface between the base **24** and the canister **10**, for example, a neck of the canister **10** that may be equipped with male threads for threading into female threads provided on the inner sidewall **54** of the base **24**. The boss **56** acts as a stop when threading or otherwise inserting a complementary portion of the canister **10** into the base **24** of the air horn **20**, thereby preventing over-threading as well as creating an air-tight seal between the air horn **20** and canister **10**. The boss **56** surrounds an inlet port **58** located within the cavity **62**. The inlet port **58** is configured to provide a flow path for the compressed gas exiting the canister **10** into an internal cavity **34** of the main body **22** (FIGS. **5** and **6**).

[0023] FIG. **4** schematically represents an upper side **42** of the air horn **20** and the actuator housing **30** protruding from the main body **22**. The actuator **32** is removed to expose an interior cavity **76** of the actuator housing **30**. The actuator housing **30** has an annular or tubular-shaped wall **50** that defines an outer sidewall **72** at the exterior of the actuator housing **30**, an inner sidewall **74** that surrounds the cavity **76**, and a lip or rim **75** therebetween at a distal end of the actuator housing **30**. The actuator housing **30** includes a first slot **84** and a second slot **86** that are defined in the rim **75** and extend axially into the wall **50**. The second slot **86** axially extends a distance from the rim **75** into the wall **50** that is greater than the distance that the first slot **84** axially extends from the rim **75** into the wall **50**. The first slot **84** is located in the rim **75** at a first rotational position corresponding to an OFF position of the actuator **32**, whereas the second slot **86** is located in the rim **75** at a second rotational position corresponding to an ON position of the actuator **32**. As depicted, the slots **84** and **86** and the ON and OFF positions defined thereby are approximately ninety degrees apart from each other, though other angular positions are foreseeable.

[0024] Located within the cavity **76** are standoffs **78** that extend up through the cavity **76**. As depicted, four standoffs **78** are provided to form an interrupted (discontinuous) peripheral wall that surrounds an annular-shaped recess **88** and a pin **90** that is centrally located within the recess **88**. As discussed below, the pin **90** is adapted to axially reciprocate within a bore **98** within the main body **22** (FIG. **10**) for the purpose of interacting with a valve assembly (not shown) conventionally

provided on canisters of the type represented in FIGS. 1 and 2, such that translation of the pin **90** toward the canister **10** and into engagement with the valve assembly of the canister **10** releases pressurized gas from the canister **10**. The standoffs **78** have arcuate cross-sections so as to have a concave surface facing the pin **90** and a convex surface facing away from the pin **90**. First and second sets of gaps **80A** and **80B** are defined by and between the standoffs **78**. The first set of gaps **80A** are disposed diametrically opposite each other, and the second set of gaps **80B** are disposed diametrically opposite each other and angularly about ninety degrees offset from the first set of gaps **80A**. As more fully described below, the actuator housing **30** and its standoffs **78** and slots **84** and **86** are configured to prevent unintentional actuation of the actuator **32**, thereby preventing the air horn **20** from producing an audible sound when the actuator **32** is intentionally actuated.

[0025] FIGS. 5 and 6 schematically depict the main body **22** of the air horn **20** and internal components thereof. Referring to FIG. 5, threads **36** are formed on a portion of the main body **22** to allow the diaphragm housing **26** to be threadably attached to the main body **22**. As can be seen in FIGS. 5 and 6, a channel wall **38** is disposed within the cavity **34** of the main body **22**. As depicted, the channel wall **38** has an annular or tubular shape that surrounds and defines a channel **44** that is fluidically connected to the bell **28**, thereby allowing air to flow through the channel **44** and into the bell **28**. Referring to FIG. 6, the tubular-shaped wall **48** of the base **24** is shown as extending into the cavity **34** of the main body **22** so as to be fluidically connected to the cavity **34** of the main body **22**. An orifice **46** is formed in and through the wall **48** of the base **24** within the cavity **34** so as to fluidically connect the inlet port **58** of the base **24** to the cavity **34** of the main body **22**, such that compressed gas entering the inlet port **58** of the base **24** from the container **10** flows into the cavity **34**. Testing has indicated that the physical relationship of the orifice **46** with the catenoid shape of the bell **28** can be tailored to result in a higher decibel level of sound as compared to an air horn utilizing the catenoid shape of the bell **28** alone or a bell with a conical shape. As represented by the broken arrow in FIG. 6, the compressed gas that enters the cavity **34** flows through the diaphragm housing **26** (not shown) before entering the channel **44**.

[0026] FIGS. 7 and 8 schematically depict detailed views of the actuator housing **30** and actuator **32** of the air horn **20**. As shown in FIG. 7, the actuator **32** includes a tubular-shaped shaft **64** with a head **66** at an upper end thereof. The shaft **64** extends into the cavity **76** of the actuator housing **30** and is sized to snugly fit within the inner sidewall **74** of the actuator housing **32**, but still able to axially reciprocate within the cavity **76** of the actuator housing **32**. The shaft **64** includes a locking feature **70** that extends radially outward from the shaft **64** and axially extends along a length of the shaft **64**. In FIG. 7, the locking feature **70** is shown as received in the first slot **84** in the wall **50** of the actuator housing **30**. As discussed below, the actuator **32** is able to be rotated so that the locking feature **70** is aligned with and receivable within the second slot **86** in the wall **50**, thereby repositioning the actuator **32** from the OFF position to the ON position. The head **66** includes a finger recess **68** that is configured as a depression in the head **66** and sized to receive a finger of a user to provide a tactile indication to the user that their finger is properly positioned on the actuator **32**. The head **66** may also include indicia, such as the arrow shown, to visually indicate the location of the locking feature **70** on the actuator **32**.

[0027] Referring to FIG. 8, the actuator **32** is depicted as translucent in order to reveal internal structures of the actuator **32**, including standoff guides **94** that axially extend within the shaft **64** of the actuator **32** from the head **66** of the actuator **32**. The standoff guides **94** are represented as two C-shaped bores separated by studs **96** therebetween that extend approximately the length of the shaft **64**. Each of the standoff guides **94** is configured to accept two of the standoffs **78** within the cavity **76** of the actuator housing **30**, and the studs **96** are configured to reside within either the first or second set of gaps **80A** or **80B** defined by and between the standoffs **78**, depending on the rotational orientation of the actuator **32**. FIGS. 7 and 8 schematically depict the instance where the actuator **32** is placed in the OFF position. In this instance, the studs **96** reside in the first set of gaps **80A** and the locking feature **70** resides within the first slot **84** in the wall **50** of the actuator housing

30. In this configuration, the actuator **32** is unable to be depressed by a user due to the shallower depth of the first slot **84** relative to the rim **75** of the wall **50**, thereby preventing the translation of the pin **90** and unintentionally sounding the air horn **10**. However, by rotating the actuator **32** ninety degrees so that the locking feature **70** is aligned with the deeper second slot **86** as shown in FIG. **9**, the actuator **32** is able to be depressed by a user to translate the pin **90** and release the compressed gas from the canister **10**, thereby generating sound with the air horn **20**.

[0028] FIG. **8** further represents a spring **92** disposed within the recess **88** surrounded by the standoffs **78** so that the spring **92** is disposed within the cavity **76** of the actuator housing **30**, surrounds the pin **90**, and is surrounded by the standoffs **78**. The spring **92** is coupled to the actuator **32** and to the cavity **76** so as to be operable to retain the locking feature **70** of the actuator **32** in an angular position aligned with the first slot **84**, but allows the actuator **32** to be axially raised and rotated so that the locking feature **70** can be rotated to an angular position that aligns the locking feature **70** with the second slot **86**, as represented in FIGS. **9** and **10**.

[0029] FIG. **10** depicts the bore **98** through which the pin **90** extends through the channel **44** surrounded by the channel wall **38** and into the inlet port **58** of the base **24** as shown in FIG. **3**, so that the pin **90** is able to interact with the valve assembly of the canister **10**. As with FIG. **9**, FIG. **10** depicts an instance in which the actuator **32** has been placed in the ON position so that the locking feature **70** is aligned with the deeper second slot **86** in the wall **50** of the actuator housing **30** to enable the actuator **32** to be depressed to actuate the air horn **20**. In this rotational position, the studs **96** (not shown in FIG. **10**) reside in the second set of gaps **80B**, and the actuator **32** is able to compress the spring **92** and translate the pin **90** toward the canister **10** and into engagement with the valve assembly (not shown) of the canister **10** to release pressurized gas from the canister **10**, which then flows into the channel **44**, through the diaphragm housing **26**, and finally through the cavity **34** of the main body **22** before exiting through the bell **28**. When the actuator **32** is released, the spring **92** returns the actuator **32** to a neutral position in which the pin **90** disengages the valve assembly of the canister **10** to terminate the release of pressurized gas from the canister **10**.

[0030] The interaction between the standoffs **78** and the standoff guides **94** as well as the studs **96** and the gaps **80A** and **80B** function to prevent the actuator **32** from being unintentionally actuated and sounding the air horn **20** in at least two ways. First, the interactions between the standoffs **78** and the standoff guides **94** and the studs **96** and the gaps **80A** and **80B** only allow the actuator **32** axial movement when in the ON or OFF positions and not at any point rotationally therebetween. Second, the mating interaction of the standoffs **78** with the standoff guides **94** function to preclude the actuator **32** from moving in any direction other than axially within the actuator housing **32**. This single axial movement eliminates the possibility of the actuator **32** engaging the pin **90** and sounding the air horn **20** when a force is applied to the actuator **32** in a direction other than vertical, i.e., along the axis of the actuator **32**.

[0031] As previously noted above, though the foregoing detailed description describes certain aspects of one or more particular embodiments of the invention, alternatives could be adopted by one skilled in the art. For example, the door finishing system and its components could differ in appearance and construction from the embodiments described herein and shown in the drawings, functions of certain components of the door finishing system could be performed by components of different construction but capable of a similar (though not necessarily equivalent) function, and various materials could be used in the fabrication of the door finishing system and/or its components. As such, and again as was previously noted, it should be understood that the invention is not necessarily limited to any particular embodiment described herein or illustrated in the drawings.

Claims

- 1.** An air horn comprising: a main body having a cavity, a channel wall within the cavity, and a channel defined and surrounded by the channel wall; a base having a wall extending into the main body, an inlet port of the air horn defined by the wall, an orifice disposed in the wall to fluidically connect the inlet port with the cavity of the main body, and connecting means for releasably connecting the air horn to a container of compressed gas; a diaphragm housing connected a first end of the main body, the diaphragm housing being fluidically connected to the cavity and the channel of the main body; a bell fluidly connected to the channel of the main body; an actuator housing having a wall connected to the main body, a cavity surrounded by the wall of the actuator housing, standoffs within the cavity of the actuator housing that define a first set of gaps between a first pair of the standoffs and a second set of gaps between a second pair of the standoffs; a pin surrounded by the standoffs and adapted to axially reciprocate within a bore within the main body; and an actuator coupled to the actuator housing so as to have rotational orientations that include a first rotational position and in a second rotational position, the actuator comprising a shaft received in the cavity of the actuator housing and standoff guides defined within the shaft that individually receive pairs of the standoffs, the standoffs and the standoff guides interacting to only permit axial movement of the actuator relative to the actuator housing when the actuator is in the first rotational position and in the second rotational position and not at any rotational position therebetween, the first rotational position corresponding to an OFF position of the actuator and the second rotational position corresponding to an ON position of the actuator.
 - 2.** The air horn of claim 1, wherein the standoff guides comprise two C-shaped bores separated by studs therebetween, the studs being configured to be able to be selectively received within the first set of gaps or within the second set of gaps, depending on the rotational orientation of the actuator.
 - 3.** The air horn of claim 1, further comprising: first and second slots axially extending in the wall of the actuator housing from a rim of the wall, the second slot axially extending a distance from the rim into the wall that is greater than a distance that the first slot axially extends from the rim into the wall, the first slot being located in the rim at the first rotational position and the second slot being located in the rim at the second rotational position; and a locking feature that extends radially outward from the shaft of the actuator and axially extends along a length of the shaft, the locking feature being selectively receivable in the first and second slots by rotating the actuator.
 - 4.** The air horn of claim 3, further comprising a spring disposed in a recess surrounded by the standoffs, the spring being coupled to the actuator and to the cavity of the actuator housing so as to be operable to retain the locking feature in a first angular position aligned with the first slot, and allow the actuator to be axially raised and rotated so that the locking feature is in a second angular position that aligns the locking feature with the second slot.
 - 5.** The air horn of claim 1, wherein the bell has a catenoid shape.
 - 6.** A method of operating the air horn of claim 1, the method comprising: releasably connecting the air horn with the connecting means to a container of compressed gas; positioning the actuator in the first rotational position to prevent actuation of the actuator and prevent release of the compressed gas from the container; and rotating the actuator to the second rotational position to permit actuation of the actuator and permit release of the compressed gas from the container.
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